Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Python Fundamentals Masterclass

Course Code: C179

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	4 October 2026	First release of the Python Fundamentals Masterclass courseware — 2-day schedule (9:30am–5:30pm) with 21 hands-on activities across six topics.	Dr Alfred Ang

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

Course Information **4**
Learning Outcomes **4**
Course Schedule **4**
Day 1 — Python Basics — Data Types, Operators, Control Structures 4
Day 2 — Functions, Modules & Packages 5
Activity Reference **6**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Course Information

Course Title	Python Fundamentals Masterclass
Course Code	C179
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch)
Mode	Instructor-led, hands-on Python coding activities (Google Colab / IDE)
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply Python programming to meet business requirements
- LO2: Code data types and operators
- LO3: Organize codes using conditional and loop for problem-solving
- LO4: Code Python functions and scripts for business use cases
- LO5: Code, import and prepare finance data using Python packages
- LO6: Test Python codes with aggregation and visualization
- LO7: Improve and document Python codes to analyse finance data

Course Schedule

Slide references [Slide n] point into the course deck Python Fundamentals Masterclass-v1.0.pptx so the trainer can navigate directly.

Day 1 — Python Basics — Data Types, Operators, Control Structures

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, learning outcomes and course outline
10:00–11:00	60 min	Topic 1 — Get Started on Python: overview, install Python, Python IDE, first script, comments [Slides 10–22]. Hands-on: Activity 1: Your First Python Script [Slide 20]
11:00–12:45	105 min	Topic 2 — Data Types: numbers, formatting, strings [Slides 23–51]. Hands-on: Activity 2:

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		Format Numbers [Slide 31]; Activity 3: Format String [Slide 36]; Activity 4: Split and Join Strings [Slide 39]
12:45–13:15	30 min	Lunch break
13:15–14:45	90 min	Topic 2 (continued) — list, tuple, dictionary, set. Hands-on: Activity 5: List Operations [Slide 43]; Activity 6: Set — Count Unique Vocabulary [Slide 48]
14:45–15:30	45 min	Topic 3 — Operators: arithmetic, compound, comparison, membership, logical [Slides 52–59]. Hands-on: Activity 7: Operators [Slide 57]
15:30–17:15	105 min	Topic 4 — Control Structures, Loop and Comprehension: if/elif/else, ternary, while and for loops [Slides 60–82]. Hands-on: Activity 8: Conditional — Income Classifier [Slide 64]; Activity 9: Fibonacci Sequence & Golden Ratio (While Loop) [Slide 69]
17:15–17:30	15 min	Day 1 recap and Q&A

9:30am – 5:30pm · 450 minutes (7.5 instructional hours), with a 30-minute lunch.

Day 2 — Functions, Modules & Packages

Time	Duration	Topic / Activity
9:30–9:45	15 min	Recap quiz on Day 1 topics
9:45–11:20	95 min	Topic 4 (continued) — for loop and range, enumerate, zip, break/continue, comprehension. Hands-on: Activity 10: For Loop — Square Numbers [Slide 72]; Activity 11: Loop Multiple Sequences with zip() [Slide 76]; Activity 12: List Comprehension [Slide 80]
11:20–13:35	135 min	Topic 5 — Function: syntax, return values,

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		default/named/variable arguments [Slides 85–106]. Hands-on: Activity 13: Function — Future Value Calculator [Slide 88]; Activity 14: Multiple Return Values — Discount & Tax [Slide 91]; Activity 15: Variable Arguments — Minimum Finder [Slide 95]
13:35–14:05	30 min	Lunch break
14:05–15:40	95 min	Topic 5 (continued) — lambda, map, filter. Hands-on: Activity 16: Lambda Function [Slide 98]; Activity 17: Map [Slide 101]; Activity 18: Filter [Slide 104]
15:40–17:15	95 min	Topic 6 — Modules & Packages: import, standard library, pip, pandas, matplotlib, yfinance [Slides 107–123]. Hands-on: Activity 19: Install Third-Party Packages [Slide 113]; Activity 20: Import & Visualize Stock Data from CSV [Slide 117]; Activity 21: Import Data from Yahoo Finance [Slide 120]
17:15–17:30	15 min	Course summary, course feedback and Q&A. End of class

9:30am – 5:30pm · 450 minutes (7.5 instructional hours), with a 30-minute lunch.

Activity Reference

Topic	Topic Focus	Activities
Topic 01: Get Started on Python	Overview · install · IDE · first script · comments	Activity 1 [Slide 20]
Topic 02: Data Types	Number · String · List · Tuple · Dictionary · Set	Activity 2 [Slide 31], Activity 3 [Slide 36], Activity 4 [Slide 39], Activity 5 [Slide 43], Activity 6 [Slide 48]
Topic 03: Operators	Arithmetic · compound · comparison · membership · logical	Activity 7 [Slide 57]
Topic 04: Control Structures,	Conditional · loops ·	Activity 8 [Slide 64],

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Loop and Comprehension	multiple sequences · comprehension	Activity 9 [Slide 69], Activity 10 [Slide 72], Activity 11 [Slide 76], Activity 12 [Slide 80]
Topic 05: Function	Syntax · returns · arguments · lambda, map, filter	Activity 13 [Slide 88], Activity 14 [Slide 91], Activity 15 [Slide 95], Activity 16 [Slide 98], Activity 17 [Slide 101], Activity 18 [Slide 104]
Topic 06: Modules & Packages	Standard library · pip · pandas, matplotlib, yfinance	Activity 19 [Slide 113], Activity 20 [Slide 117], Activity 21 [Slide 120]

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg